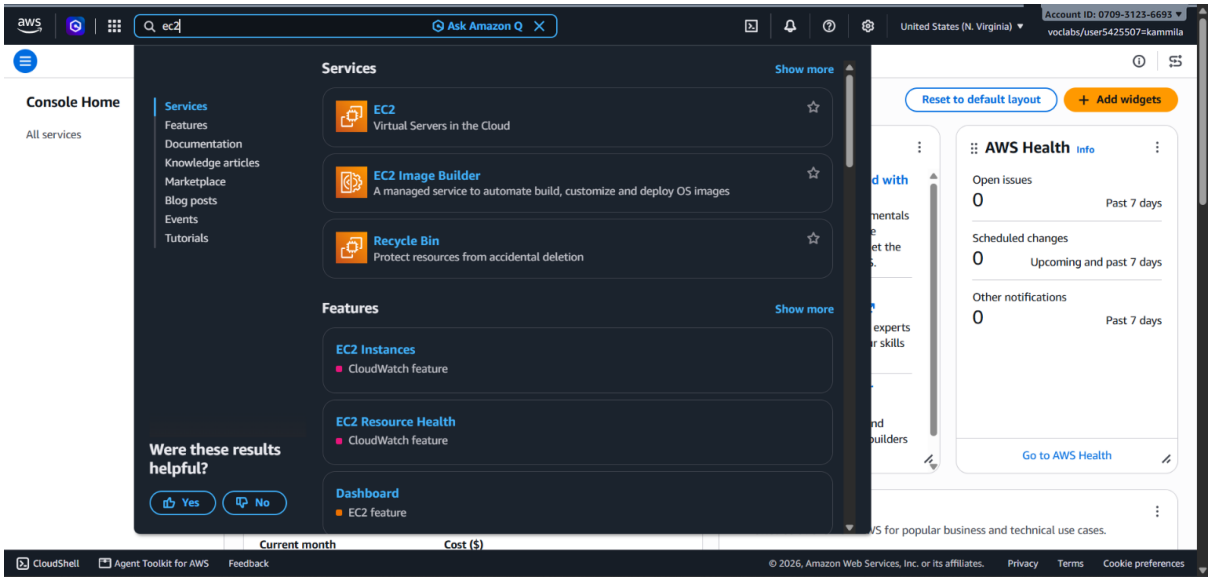
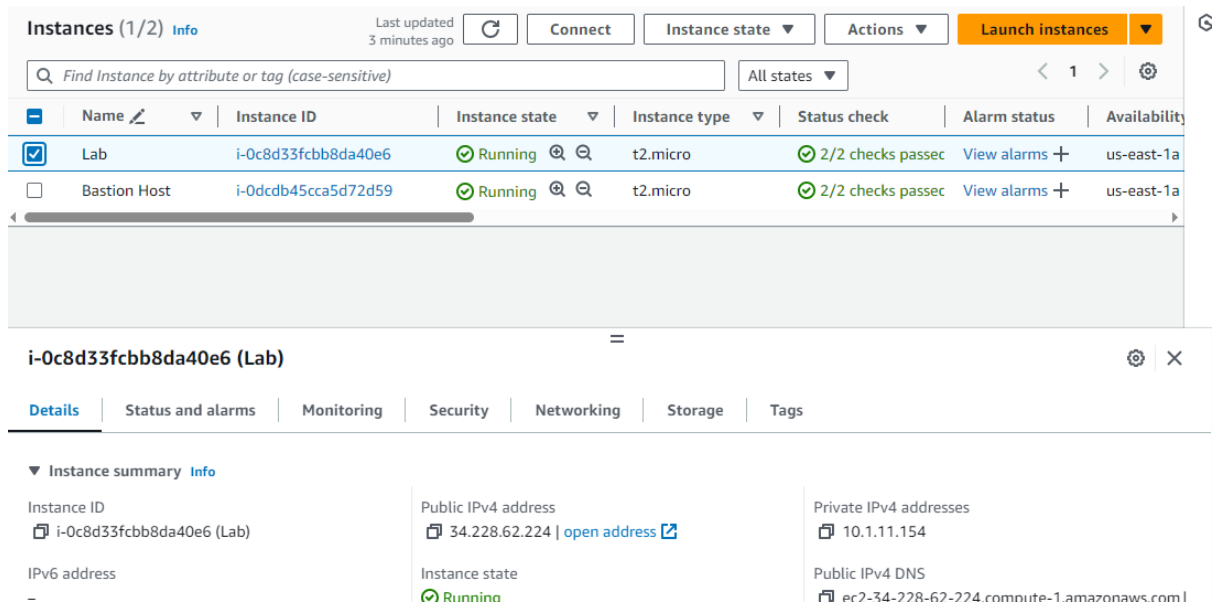
Practical 4

4. Working with Amazon Elastic Block Store (EBS)

Step 1: In the AWS Management Console, in the search box next to service, search for and select EC2**.**
In the left navigation pane, choose **Instances**.
An Amazon EC2 instance named **Lab** has already been launched for lab.



Note the **Availability Zone** of the instance. It will look similar to *us-east-1a*.
In the left navigation pane, choose **Volumes**.
You will see an existing volume that is being used by the Amazon EC2 instance. This volume has a size of 8 GiB, which makes it easy to distinguish from the volume you will create next, which will be 1 GiB in size.

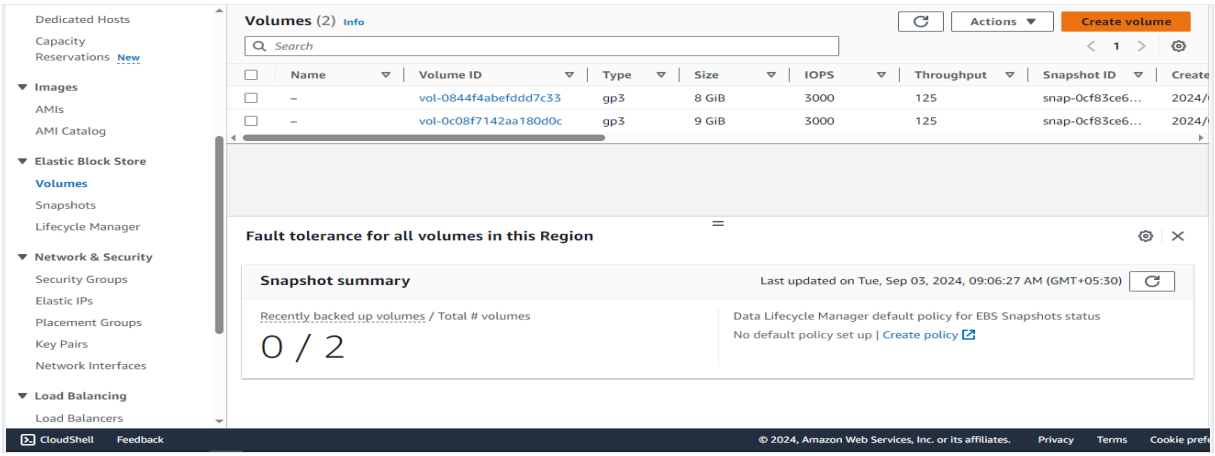


Step 2: Choose create volume then configure:

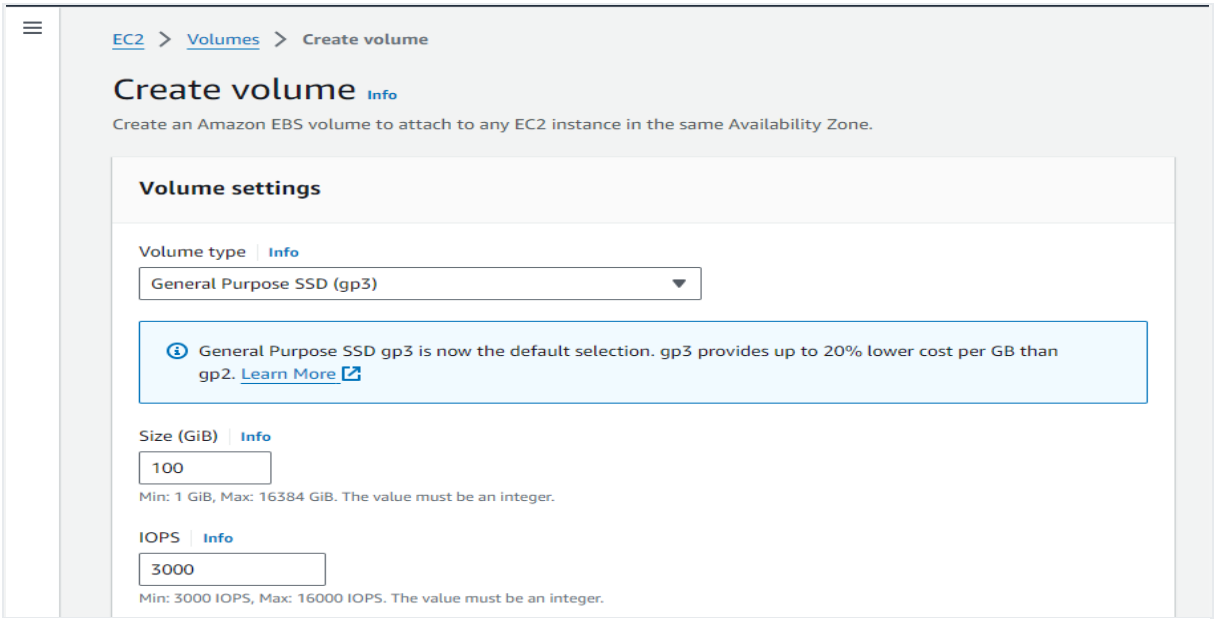
Volume Type: *General Purpose SSD (gp2)*

Size (GiB): 1. **NOTE**: You may be restricted from creating large volumes.

Availability Zone: Select the same availability zone as EC2 instance.



Choose **Add tag**
In the Tag Editor, enter:
Key: Name
Value: My Volume



Choose Create a volume

new volume will appear in the list, and will move from the *Creating* state to the *Available* state. You may need to choose **refresh** to see new volume.

Step 3: Attach the Volume to an Instance

In this task you will attach the new EBS volume to the Amazon EC2 instance.

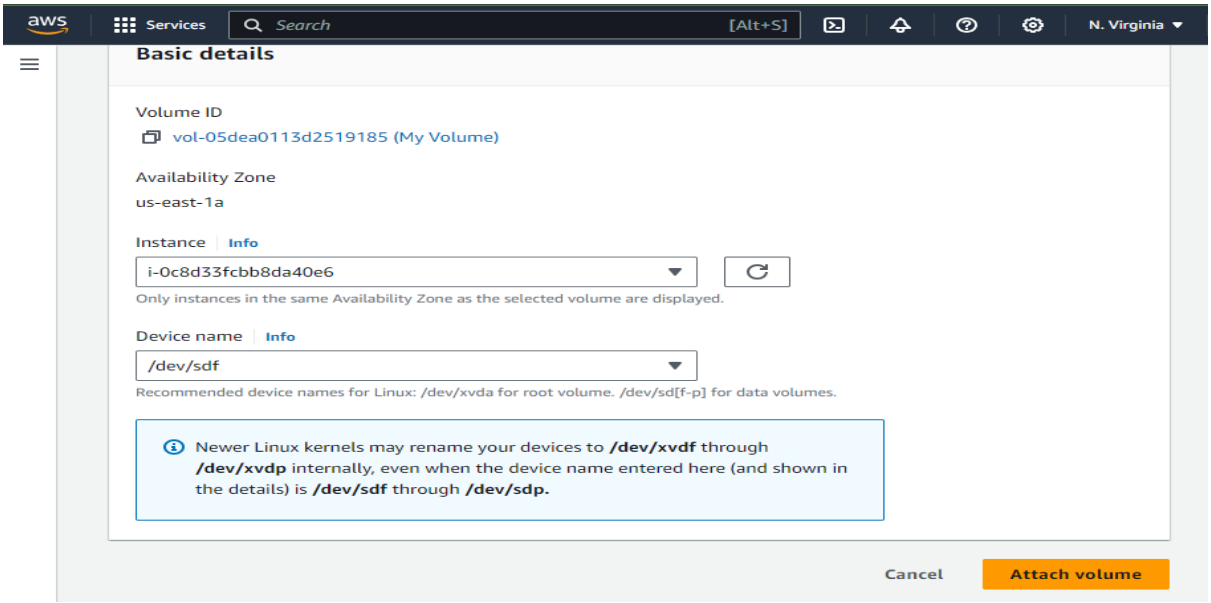
Select **My Volume**.

In the **Actions** menu, choose **Attach volume**.

Choose the **Instance** field, then select the **Lab** instance.

Choose **Attach volume**

The volume state is now *In-use*.



Step 4: Connect to Amazon EC2 Instance

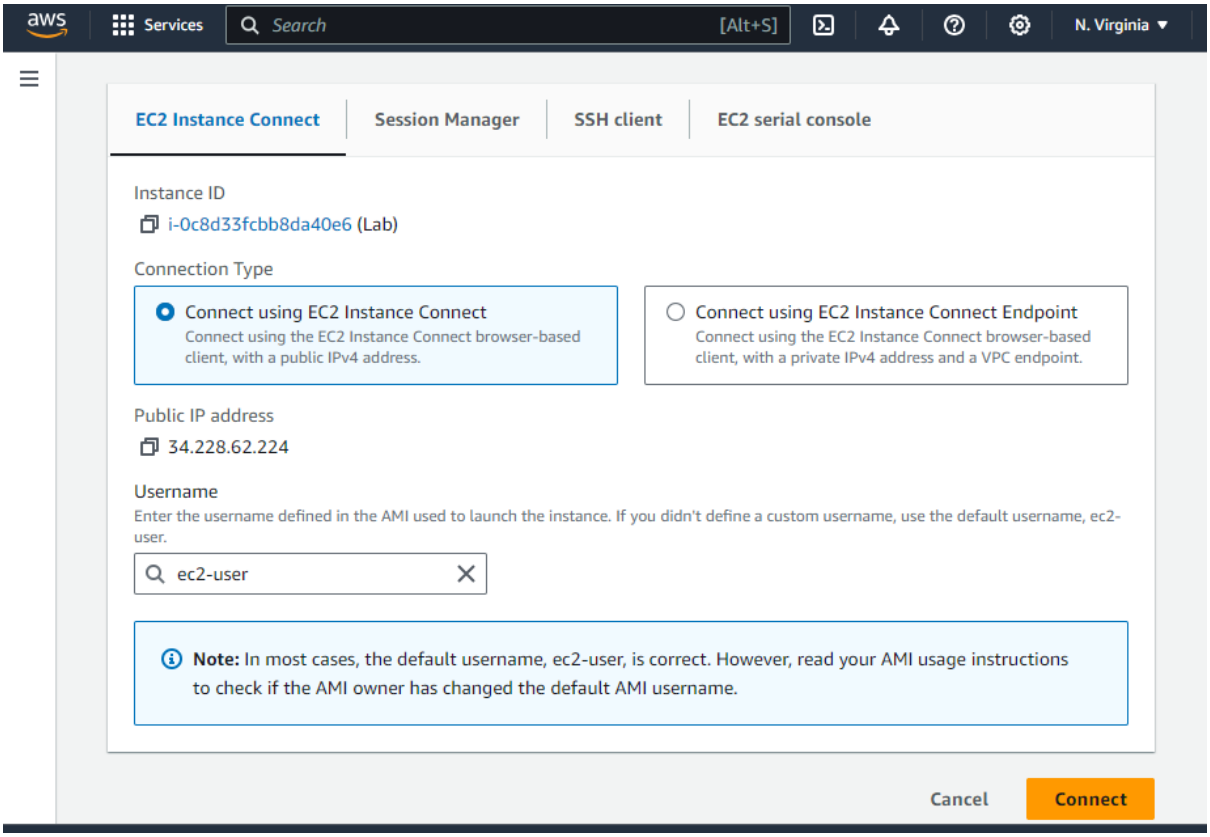
In this task, you will connect to the EC2 instance using EC2 Instance Connect which provides access to a terminal in the browser.

14. In the AWS Management Console, in the search box next to **Services** , search for and select **EC2**.

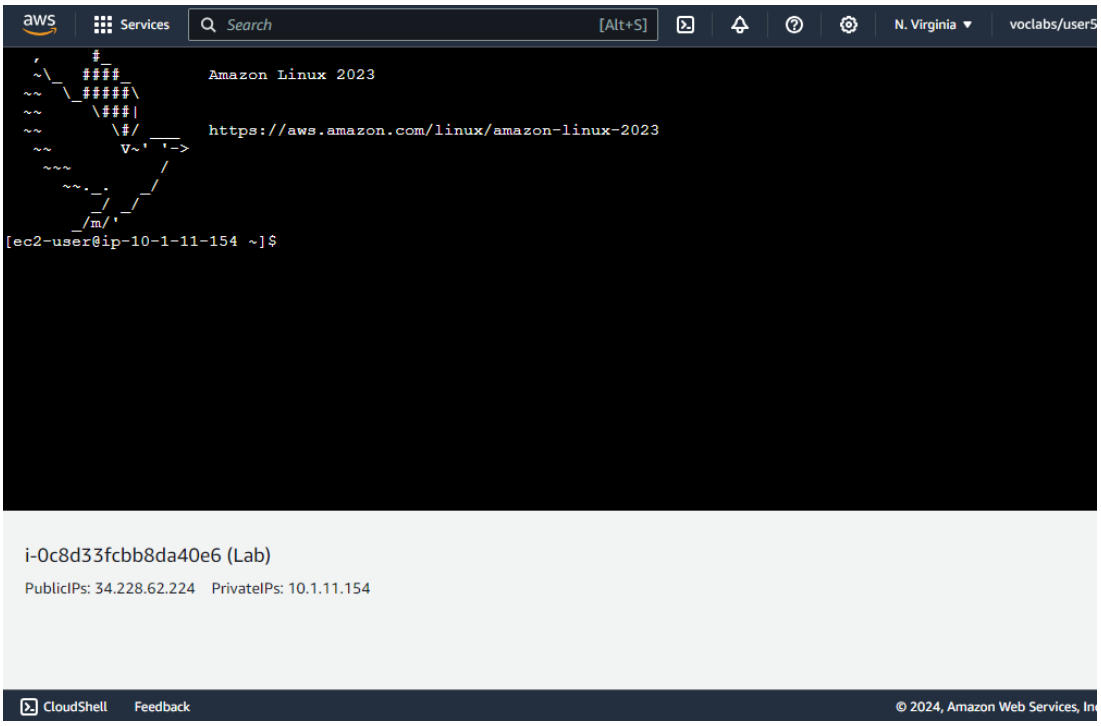
15. Choose **Instances**.

16. Select the **Lab** instance, and then choose **Connect**.

17. On the **EC2 Instance Connect** tab, choose **Connect**.



An EC2 Instance Connect terminal session opens and displays a $ promp



Step 5: Create and Configure File System
In this task, you will add the new volume to a Linux instance as an ext3 file system under the /mnt/data-store mount point.
View the storage available on instance:
Run the following command:
Dh -h

Create an ext3 file system on the new volume:

sudo mkfs -t ext3 /dev/sdf
The output should indicate that a new file system was created on the attached volume.

Create a directory for mounting the new storage volume:

sudo mkdir /mnt/data-store

Mount the new volume:

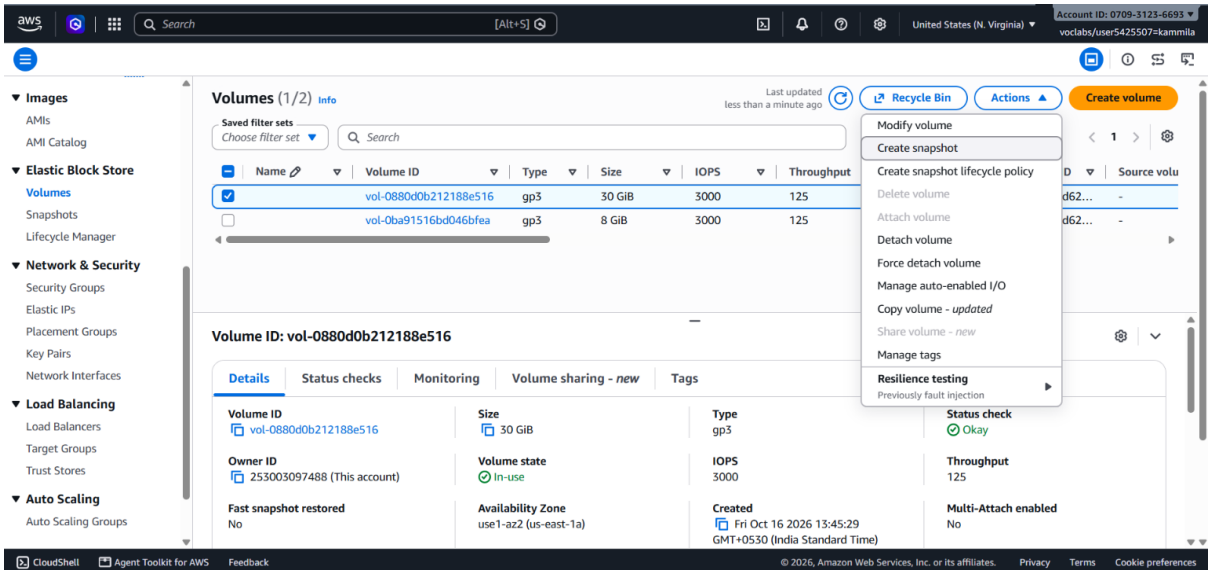
sudo mount /dev/sdf /mnt/data-store
To configure the Linux instance to mount this volume whenever the instance is started, you will need to add a line to */etc/fstab*. Run the command below to accomplish that:

echo "/dev/sdf /mnt/data-store ext3 defaults,noatime 1 2" | sudo tee -a /etc/fstab

Step 6: Create an Amazon EBS Snapshot
In this task, you will create a snapshot of EBS volume.

In the **EC2 Console**, choose **Volumes** and select **My Volume**.

In the **Actions** menu, select **Create snapshot**.



Choose **Add tag** then configure:

- **Key:** Name
- **Value:** My Snapshot
- Choose **Create snapshot**

